Rate Limiting ?

500K

1 person 500K

500K hitting 1 req

~~500K~~ 200K

1 sec → 10 Req → Person A

Rate Limit them

20 → 30 → (X)

H.L.D

Functional Reqr

↳ on what basis, should I rate limit them
↳ how identify → API-key, I.P, user-id

- limit req's → Rules →

hashmap ⇒ userid → hashmap, set

U123, t1 t2 t3 ~~t4~~ t4

(X) limit

B.E

(X) 429 ↳ too many reqs

Respond with proper msgs → 429

rate Limitter

↳ new rule

non fuctional :→

CAP → Consistency Availability

+ eventual Consistency

~~scalability~~

Latency

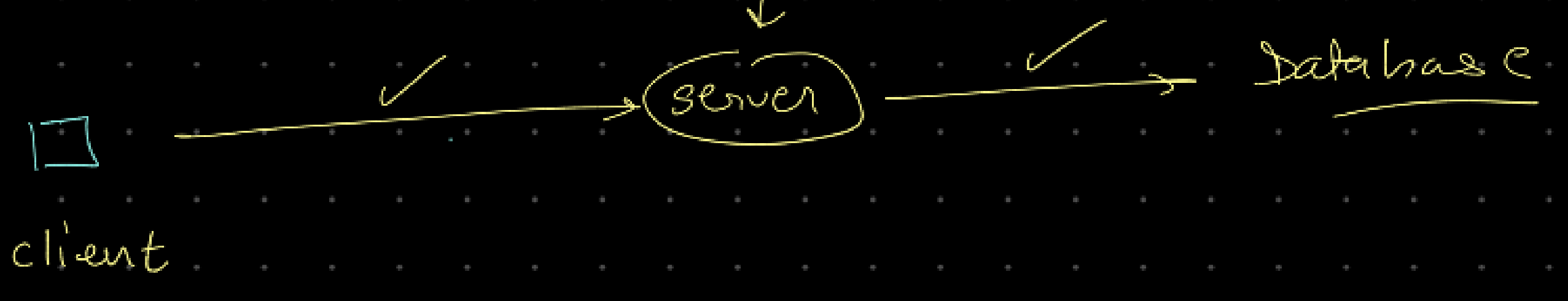
client → calls → B.E → 1 sec

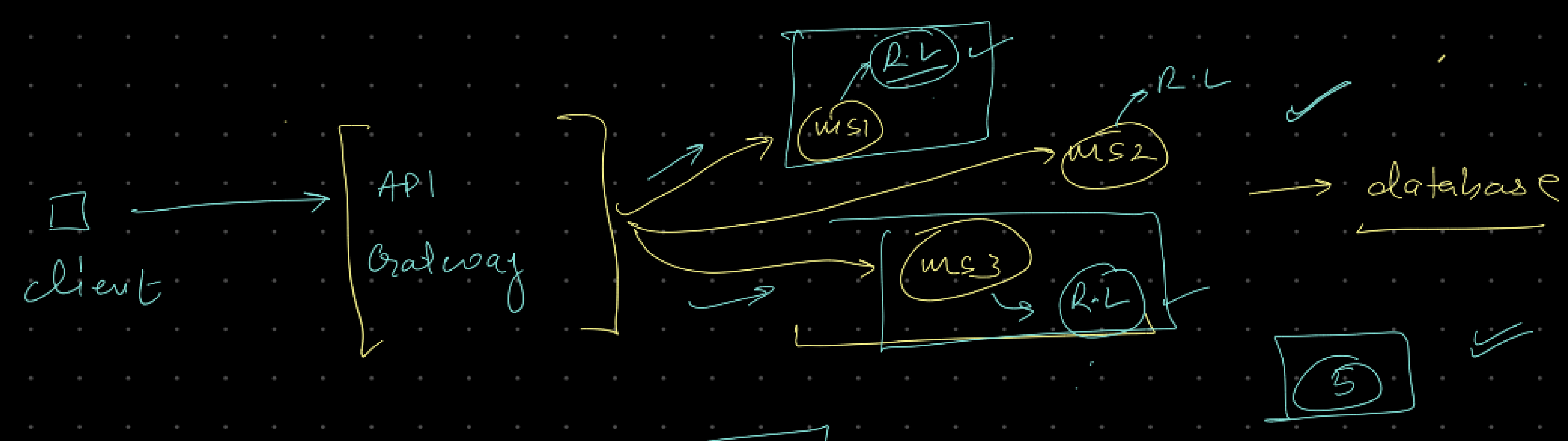
rate limit checks lightening fast

10 ms → Redis

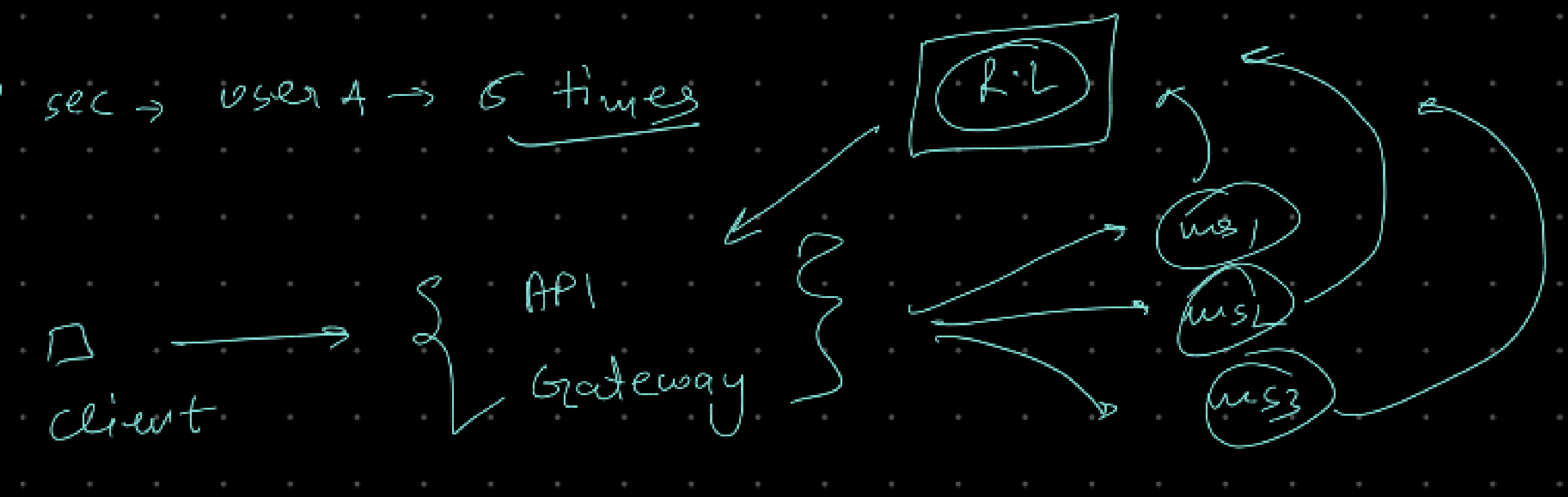
Scale :→ 100 M DAU ⇒ 1M → qps

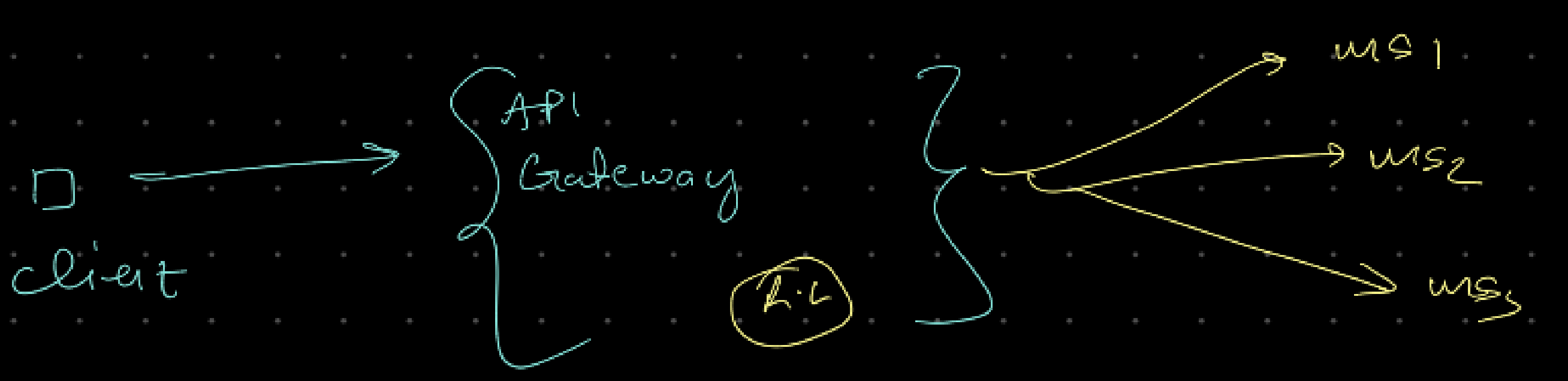
userid
I.P
API Key

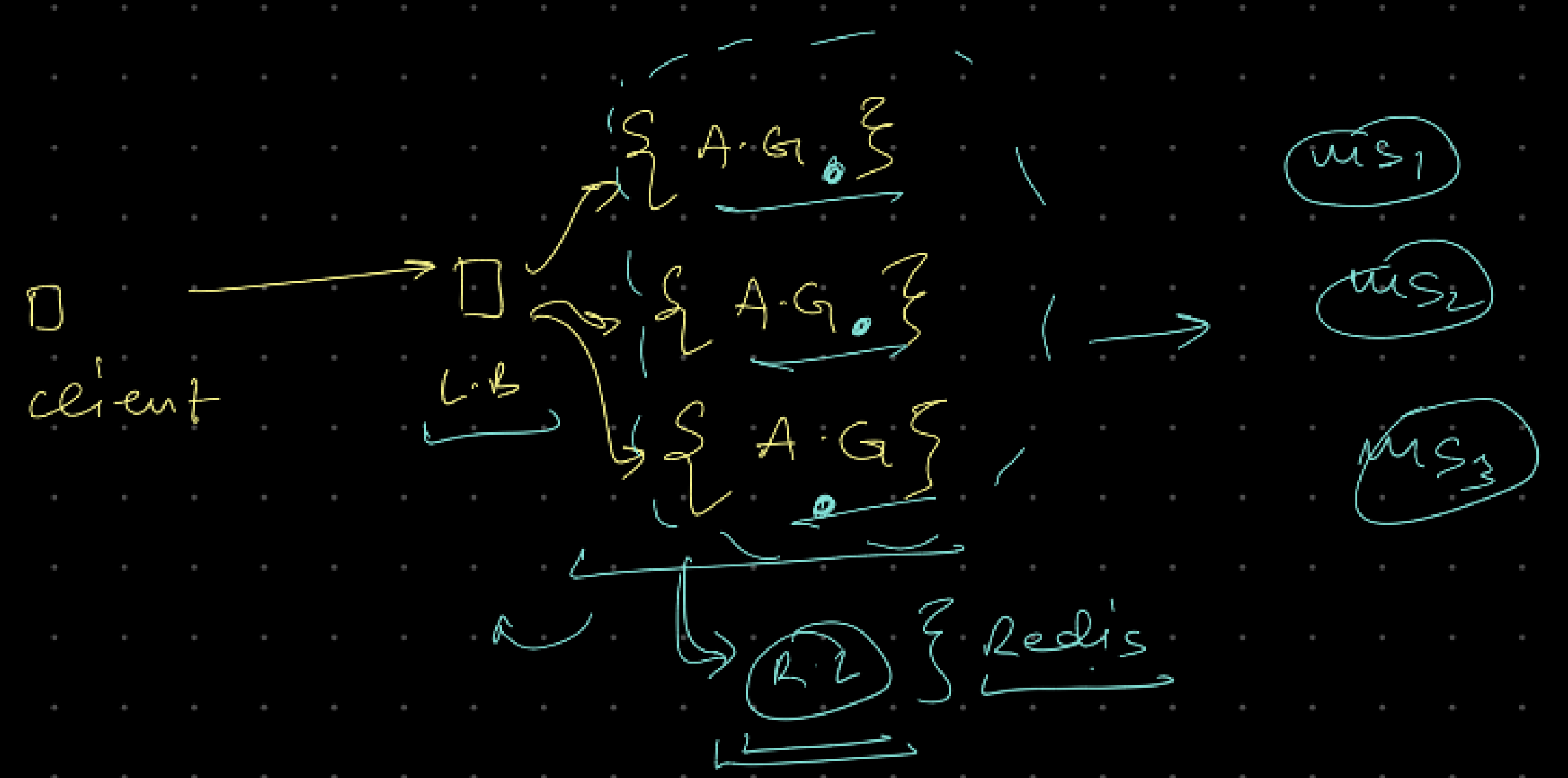




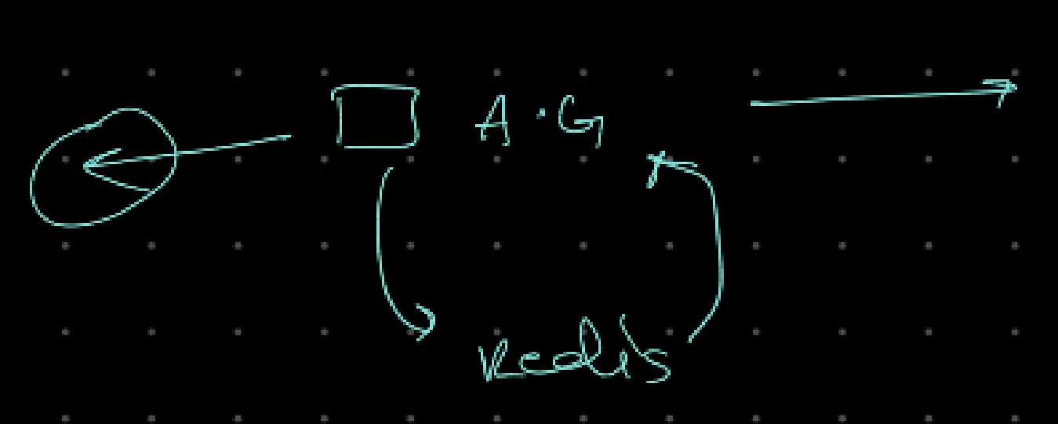
5 sec → user A → 5 times







Rate-Limit → API → header → API key
body → user id



Query Params
↳ Request headers

Response Codes ↳ 429

Response headers → headers in the response

X-RateLimit-Limit → 100
X-RateLimit-Remaing → ~~0~~ 3
X-RateLimit-Reset → 10 23 43 9 . . .

Rate Limiting ? → algorithms

token-Bucket

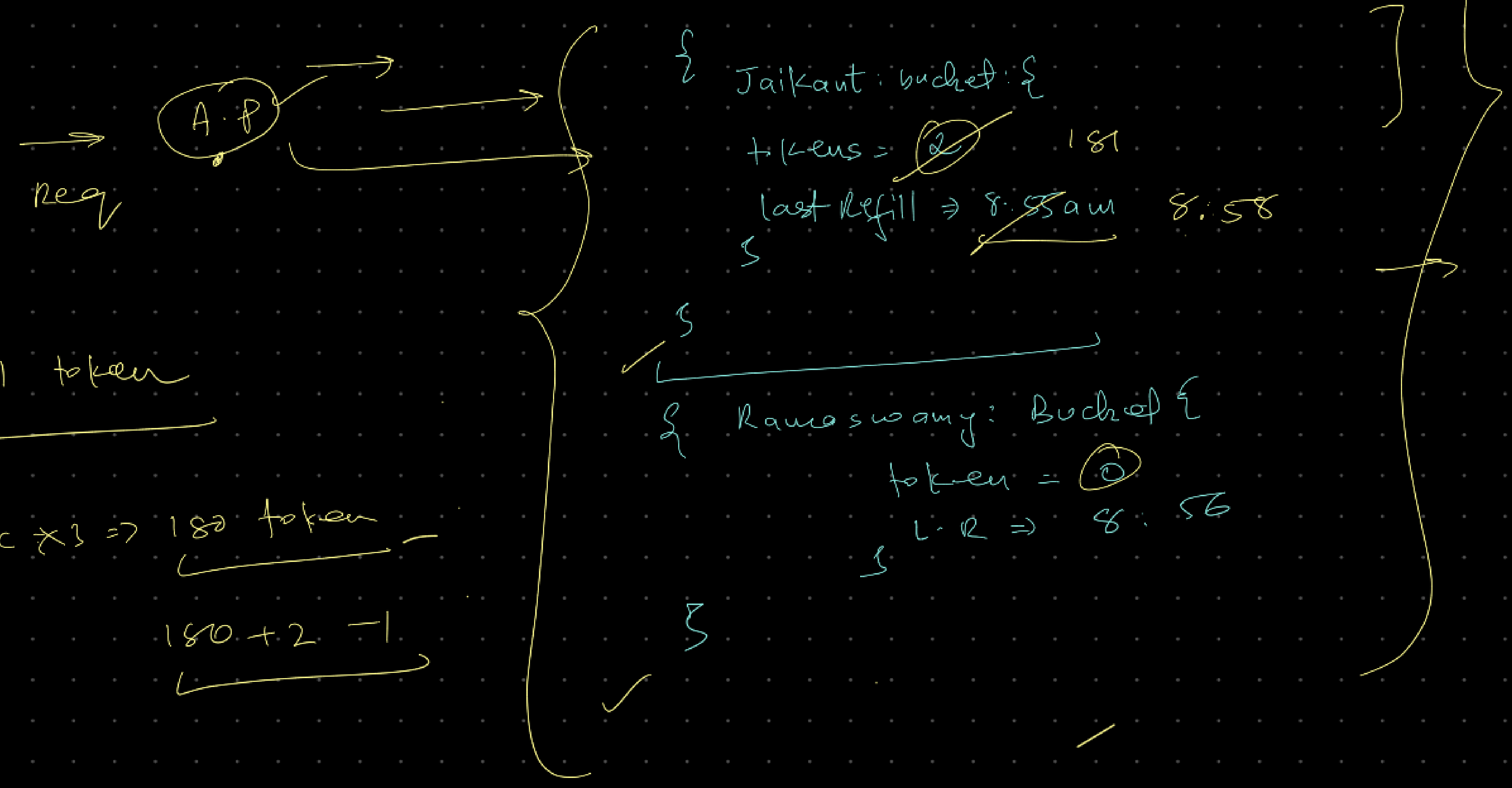
05 35 6 9
Bucket — 1 2 3 4
0 ~~1~~ 0 →

0 → 1R →
1sec → ~~R~~
100 req → 0 → 2sec → (X)
1 R → ✓
4 rd → 5 sec
103 → (X)
6 sec → 1R ✓
7 → 8 → R.L

3 sec → 1R → 0
1R → 3

→ 0 → 1

R → ✓



{
Jaikant : bucket : {
tokens = ~~2~~ 181
last Refill ⇒ ~~8:55~~am 8:58
}
}

{ Ramaswamy : Bucket {
token = 0
L.R ⇒ 8 : 56
}
}

1 sec → 1 token

60sec × 3 ⇒ 180 token

180 + 2 − 1